Chapter 31

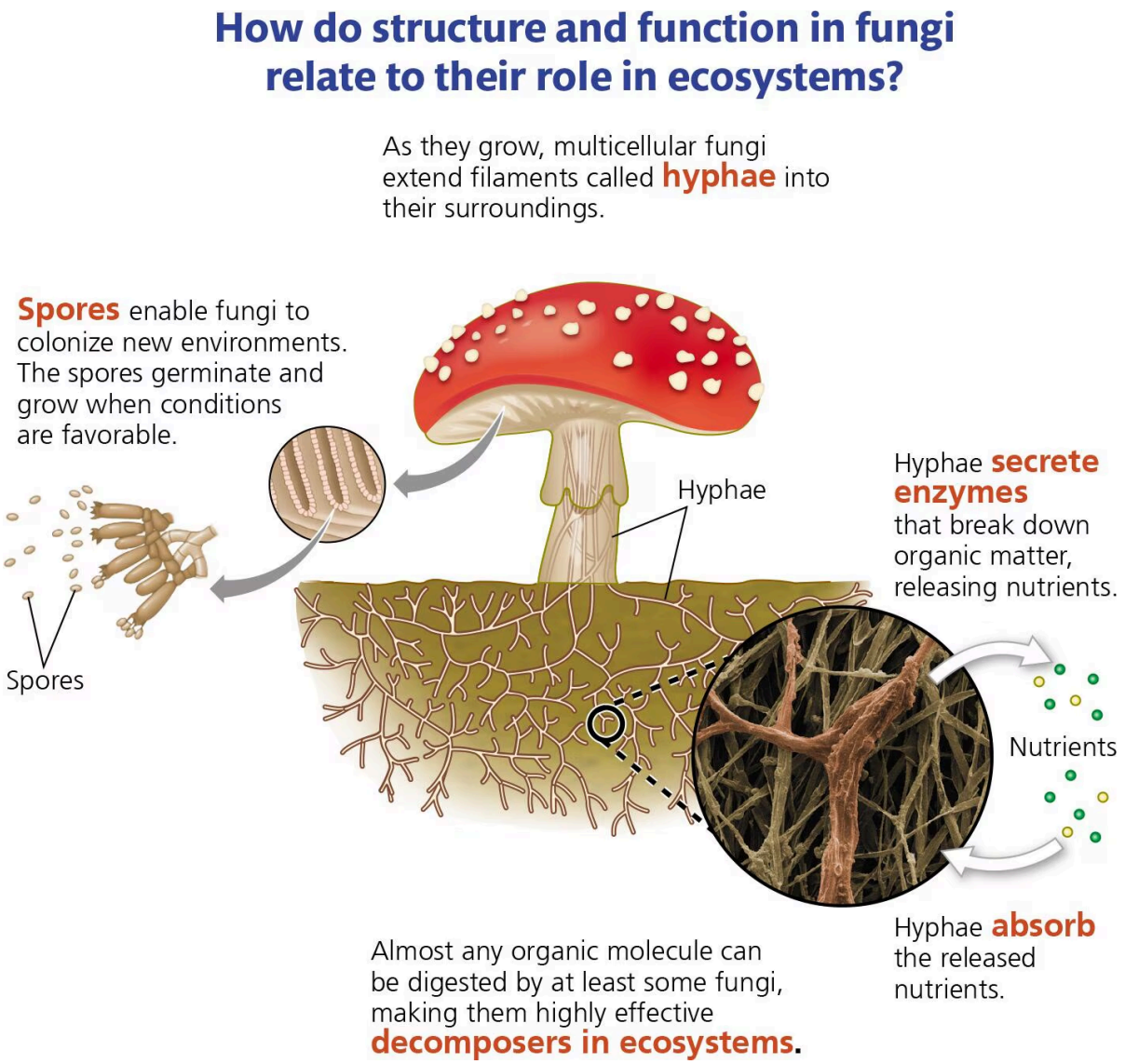
Campbell Biology, AP Edition, 12e

Chapter 31

Fungi

Figure 31.1

These little mushrooms are just the small aboveground extension of a vast network of filaments located below the forest floor. Such underground fungal networks, called mycelia, in some cases link mushrooms that are hundreds of meters apart. In fact, the largest known mycelium spreads beneath 965 hectares of forest—more than the area of 1,800 football fields.



As they grow, multicellular fungi extend filaments called hyphae into their surroundings. Spores enable fungi to colonize new environments. The spores germinate and grow when conditions are favorable. Hyphae secrete enzymes that break down organic matter, releasing nutrients. Hyphae absorb the released nutrients. Almost any organic molecule can be digested by at least some fungi, making them highly effective decomposers in ecosystems.

Key Concepts

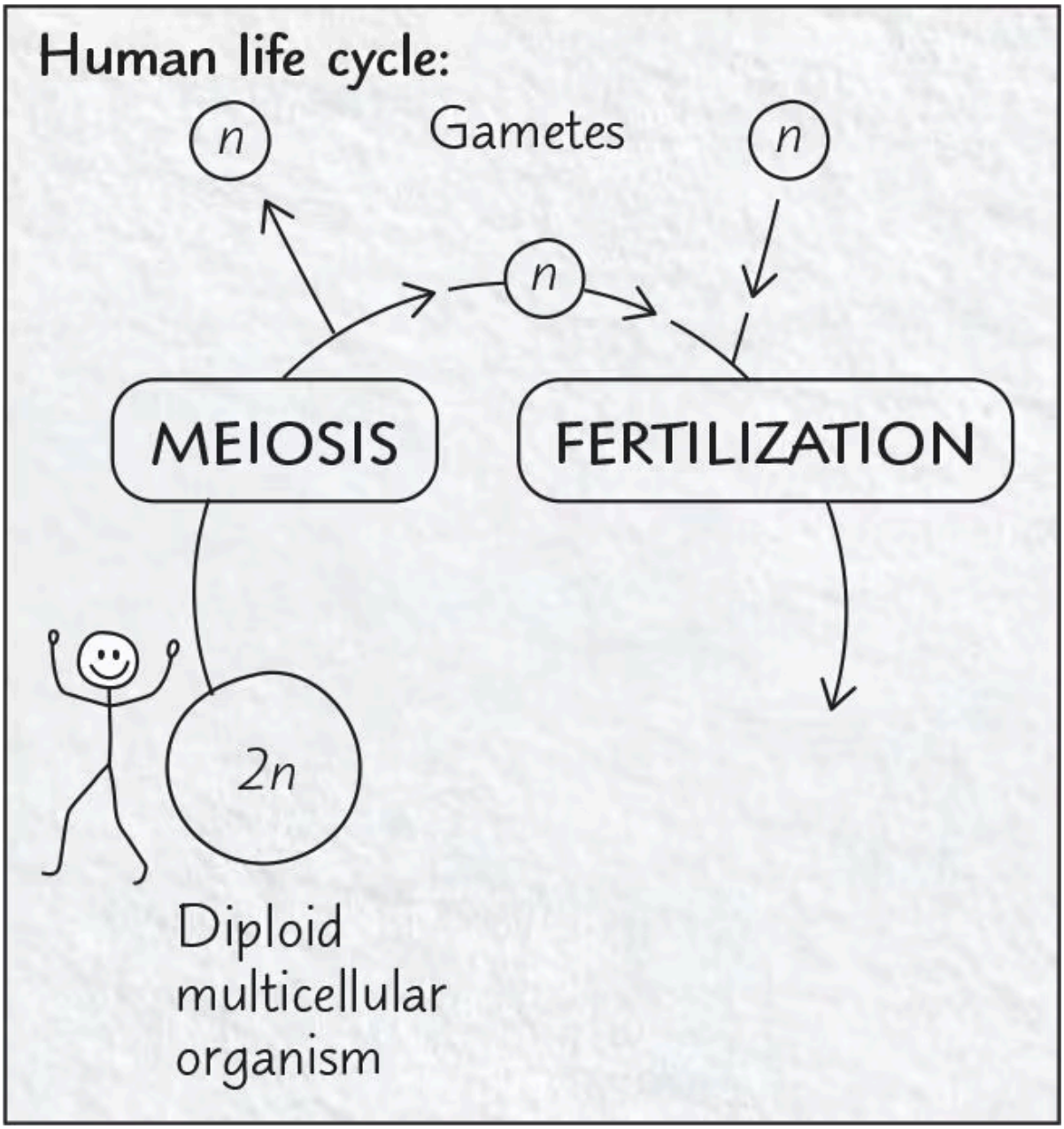
- **31.1Fungi are heterotrophs that feed by absorption****
- **31.2Fungi produce spores through sexual or asexual life cycles****
- **31.3The ancestor of fungi was an aquatic, single-celled, flagellated protist****
- **31.4Fungi have radiated into a diverse set of lineages****

- **31.5Fungi play key roles in nutrient cycling, ecological interactions, and human welfare****

Molecular evidence indicates that fungi were among the earliest colonizers of land **(BIG IDEA 3)**, successfully using their large surface area to absorb water and nutrients **(BIG IDEA 2)**. They diverged from animals over 1 billion years ago **(BIG IDEA 1)** and play a critical role in the environment through mutualistic relationships and nutrient recycling **(BIG IDEA 4)**.

Study Tip

Draw a diagram: To help you recognize key differences between the life cycles of fungi and humans, draw and label simple diagrams like the partial example shown here depicting the life stages at which meiosis, mitosis, gamete formation, a multicellular organism, and fertilization occur in humans and in fungi.



A diploid multicellular organism, depicted as a circle with 2 n and a stick figure of a human next to it. The 2 n organism undergoes meiosis and forms haploid gametes, n. One gamete, n, fuses with another gamete, n, and undergoes fertilization.

The hidden network of fungal filaments in Figure 31.1 is a fitting symbol of the neglected grandeur of the kingdom Fungi. Most of us are barely aware of these eukaryotes beyond the mushrooms we eat or the occasional brush with athlete's foot. Yet fungi are a huge and important component of the biosphere. Some fungi are exclusively single-celled, though most have complex multicellular bodies. These diverse organisms are found in just about every imaginable terrestrial and aquatic habitat.

Fungi are not only diverse and widespread but also essential for the well-being of most ecosystems. They break down organic material and recycle nutrients, allowing other organisms to assimilate essential chemical elements. In this chapter, we will investigate the structure and evolutionary history of fungi, survey the major groups of fungi, and discuss their ecological and commercial significance.

Get Ready for This Chapter

In this chapter, you will learn about the structure, evolutionary history, diversity, and ecological significance of fungi. Before beginning this chapter, you should be able to do the following:

- Explain what decomposers are and how they affect ecosystems (see Concept 1.1, Figure 1.9, and Concept 27.5).
- Distinguish between asexual and sexual reproduction (see Concept 13.1).
- Define key ecological interactions: symbiosis, commensalism, mutualism, and parasitism (see Concept 27.5).

The following questions provide a quick check of your basic knowledge in these areas.

1. Which of the following statements about decomposers is true?

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They are detrimental to ecosystems where they are found.
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That's incorrect. Try again.

Incorrect. Review the following material: Decomposers obtain organic molecules (carbon) and chemical energy by breaking down waste materials, leaf litter, and the bodies of dead organisms. When decomposers break down these substances, they return chemicals to the environment, making them available to plants and other photosynthetic organisms. Decomposers include prokaryotic and eukaryotic species.

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They are autotrophs.
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That's incorrect. Try again.

Incorrect. Review the following material: Decomposers obtain organic molecules (carbon) and chemical energy by breaking down waste materials, leaf litter, and the bodies of dead organisms. When decomposers break down these substances, they return chemicals to the environment, making them available to plants and other photosynthetic organisms. Decomposers include prokaryotic and eukaryotic species.

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They include only prokaryotic species.
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That's incorrect. Try again.

Incorrect. Review the following material: Decomposers obtain organic molecules (carbon) and chemical energy by breaking down waste materials, leaf litter, and the bodies of dead organisms. When decomposers break down these substances, they return chemicals to the environment, making them available to plants and other photosynthetic organisms. Decomposers include prokaryotic and eukaryotic species.

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They break down waste products, leaf litter, and the bodies of dead organisms.
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That's correct.

Correct! If you had trouble with this question, review the following material: Decomposers are heterotrophs, or consumers, that obtain organic molecules (carbon) and chemical energy by breaking down organic waste materials, leaf litter, and the bodies of dead organisms. When decomposers break down these substances, they return chemicals to the environment, making them available to plants and other photosynthetic organisms. Decomposers include prokaryotic and eukaryotic species.

2. In asexual reproduction,

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one parent passes copies of all of its genes to its offspring without the fusi
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That's correct.

Correct! If you had trouble with this question, review the following material: In asexual reproduction, a single individual is the sole parent and passes copies of all its genes to its offspring without the fusion of gametes; meiosis and crossing over do not occur. Single-celled eukaryotic organisms can reproduce asexually by mitotic cell division, in which DNA is copied and allocated equally to two daughter cells. Some multicellular organisms also reproduce asexually; in such organisms, the cells of the offspring arise via mitosis in the parent. Note that in asexual reproduction, each offspring is genetically identical to its parent, or nearly so (as a result of mutation, some offspring may exhibit slight genetic differences from the parent).

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one parent passes copies of all of its genes to its offspring after crossing over o
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That's incorrect. Try again.

Incorrect. Review the following material: In asexual reproduction, a single individual is the sole parent and passes copies of all its genes to its offspring without the fusion of gametes; meiosis and crossing over do not occur. Single-celled eukaryotic organisms can reproduce asexually by mitotic cell division, in which DNA is copied and allocated equally to two daughter cells. Some multicellular organisms also reproduce asexually;

in such organisms, the cells of the offspring arise via mitosis in the parent. Note that in asexual reproduction, each offspring is genetically identical to its parent, or nearly so (as a result of mutation, some offspring may exhibit slight genetic differences from the parent).

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each offspring has some cells derived by mitosis in the parent and other cells deri
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That's incorrect. Try again.

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each parent passes copies of all of its genes to its offspring, causing the offspri
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3. Suppose two species live in close contact with each other. One species benefits by eating the tissues of the other, and the other is harmed (by having its tissues consumed). The ecological interaction between these species is an example of

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   symbiosis and mutualism.
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That's incorrect. Try again.

Incorrect. Review the following material: A symbiosis is an ecological relationship in which two species live in close contact with one another. Parasitism is an ecological relationship in which one organism (the parasite) eats the tissues or body fluids of another (the host), thereby harming the host. A mutualism is an

ecological relationship in which each species benefits. A commensalism is an ecological relationship in which one species benefits, and the other is neither harmed nor helped.

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parasitism and symbiosis.
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That's correct.

Correct! If you had trouble with this question, review the following material: These two species are engaged in a symbiosis, an ecological relationship in which two species live in close contact with one another. This particular symbiosis is also an example of parasitism, an ecological relationship in which one organism (the parasite) eats the tissues or body fluids of another (the host), thereby harming the host. In other cases, the two species engaged in a symbiosis form a mutualism (an ecological relationship in which each species benefits) or a commensalism (an ecological relationship in which one species benefits and the other is not harmed or helped).

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symbiosis and commensalism.
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That's incorrect. Try again.

Incorrect. Review the following material: A symbiosis is an ecological relationship in which two species live in close contact with one another. Parasitism is an ecological relationship in which one organism (the parasite) eats the tissues or body fluids of another (the host), thereby harming the host. A mutualism is an ecological relationship in which each species benefits. A commensalism is an ecological relationship in which one species benefits, and the other is neither harmed nor helped.

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mutualism and parasitism.
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Learning Objectives

After studying this chapter, you should be able to do the following:

- 31.1Describe the structure of the fungal body.

- 31.2Compare and contrast sexual and asexual reproduction in fungi.
- 31.3Explain how fungi may have evolved from a single-celled protist.
- 31.4Identify and characterize major phylogenetic groups of fungi.
- 31.5Give examples of how fungi interact with other organisms.